

Model

1 Model

1.1 Environment

Consider a metropolitan area consisting of a finite set of locations $S = \{1, 2, \dots, n\}$. These locations are partitioned into municipalities $g \in G$:

$$S = \bigcup_{g \in G} S^g, \quad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

Each location $i \in S$ is endowed with a fixed supply of land $\bar{H}_i$, allocated between residential and commercial use by local zoning:

$$\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where $\bar{H}_i^R$ is residential floor space capacity and $\bar{H}_i^F$ is commercial floor space capacity. Both are chosen by the municipal government and assumed to be fully utilized in equilibrium. Land at each location is owned by absentee landlords who collect $q_i^R \bar{H}_i^R + q_i^F \bar{H}_i^F$ and consume it outside the metropolitan area; this rental income has no further equilibrium feedback — it does not enter household budgets or municipal welfare.

1.1.1 Transportation

Locations are connected by a transportation network (S, E) , where $E \subseteq S \times S$ is a set of directed edges (links). Transportation investment $I_{k\ell}$ is chosen at the *edge* level, for each $(k, \ell) \in E$. The utility cost of traveling through edge (k, ℓ) is

$$d_{k\ell} = \exp(\kappa \times \text{time}_{k\ell}),$$

where $\text{time}_{k\ell}$ is the travel time on edge (k, ℓ) and $\kappa > 0$ is a scale parameter, following Ahlfeldt et al., 2015. The travel time $\text{time}_{k\ell}$ is a function of the edge-level traffic $n_{k\ell}$ and investment $I_{k\ell}$:

$$\text{time}_{k\ell} = \delta_{kl}^0 + \delta_{kl}^1 \frac{n_{k\ell}^\rho}{I_{k\ell}^\beta}.$$

Following Fajgelbaum and Schaal, 2020, δ_{kl}^0 is the free-flow travel time on edge (k, ℓ) , $\delta_{kl}^1 > 0$ is a scale parameter, and $\rho, \beta > 0$ are elasticities of travel time with respect to traffic and investment respectively. Therefore for a given route r , the commuting cost τ_{ij} between residence i and workplace j is the product of the costs of all edges along that route:

$$\tau_{ijr} = \prod_{(k, \ell) \in r_{ij}} d_{k\ell}(I_{k\ell}, n_{k\ell}),$$

where r_{ij} is the route chosen by commuters from i to j .

There are three types of agents: households, competitive firms, and governments. Households choose residence and workplace locations. Firms produce a freely tradable good using labor and commercial floor space. Government decisions operate at two levels: municipal governments choose local zoning, and a metropolitan transportation authority chooses investment across all links. In the decentralized equilibrium, municipalities and the transportation authority choose policies simultaneously, each treating the other's decisions as given.

1.2 Households

A household chooses a residence location i , a workplace j , and a route r from a finite, exogenously given choice set R_{ij} of feasible routes between i and j (e.g. the k shortest GIS paths, or distinct road/transit alternatives) to maximize utility. Preferences over the numaire good Q_{ij} and residential floor space h_{ij}^R are Cobb-Douglas:

$$u_{ijr}(\omega) = \frac{B_i(N_i, L_i)}{\tau_{ijr}} \left(\frac{Q_{ij}}{\alpha} \right)^\alpha \left(\frac{h_{ij}^R}{1-\alpha} \right)^{1-\alpha} v_{ij,r}(\omega),$$

where $B_i(N_i, L_i) = \bar{B}_i / \phi(N_i, L_i) > 0$ is the effective residential amenity at location i : an exogenous location-specific level $\bar{B}_i$, discounted by the congestion disutility $\phi(N_i, L_i)$ from local residential and workplace density (defined below in §Municipal Zoning). $\alpha \in (0, 1)$ is the expenditure share on the composite good, τ_{ijr} is the commuting cost along route r , and $\{v_{ij,r}(\omega)\}$ is a vector of idiosyncratic shocks, jointly distributed according to a nested (generalized extreme value) Fréchet copula with cumulative distribution function

$$F(\{v_{ij,r}\}) = \exp \left[- \sum_{ij \in S^2} T_i E_j \left(\sum_{r \in R_{ij}} v_{ij,r}^{-\theta} \right)^{\epsilon/\theta} \right], \quad \theta \geq \epsilon > 1. \quad (1)$$

The outer exponent is $+\epsilon/\theta$, not $-\epsilon/\theta$: a valid CDF must satisfy $F \rightarrow 1$ as any $v_{ij,r} \rightarrow \infty$ holding the rest fixed, which pins the sign down. T_i, E_j enter at the outer (i, j) nest only, not inside the inner route sum: they are location-pair accessibility scales, not route-specific ones, exactly as in the single-route model. Setting $R_{ij} = \{r\}$ for every (i, j) collapses (1) to the original univariate Fréchet marginal $\Pr(v_{ij} \leq z) = \exp(-T_i E_j z^{-\epsilon})$, confirming the nesting strictly generalizes it; $\theta \geq \epsilon$ is required for (1) to be a valid (positively correlated within-nest) GEV copula.

The household's budget constraint is

$$Q_{ij} + q_i^R h_{ij}^R \leq w_j,$$

where q_i^R is the residential rent at location i and w_j is the wage at workplace j — local congestion no longer enters the budget directly; it operates entirely through the amenity $B_i(N_i, L_i)$ above. Optimal demands are $Q_{ij}^* = \alpha w_j$ and $h_{ij}^{R*} = (1-\alpha)w_j/q_i^R$, giving the deterministic component of indirect utility for a given route,

$$\tilde{V}_{ijr} = \frac{B_i(N_i, L_i) w_j}{\tau_{ijr} (q_i^R)^{1-\alpha}}.$$

Residential floor space clears when aggregate demand from all residents of i exhausts the fixed supply $\bar{H}_i^R$: summing h_{ij}^{R*} over workplaces j , weighted by the population $N\pi_{ij}$ commuting from i to j ,

$$\sum_j N\pi_{ij} \frac{(1-\alpha)w_j}{q_i^R} = \bar{H}_i^R \iff q_i^R = \frac{(1-\alpha) \sum_j N\pi_{ij} w_j}{\bar{H}_i^R}, \quad (2)$$

which pins down q_i^R as a function of the commuting-flow pattern $\{\pi_{ij}\}$, wages, and the zoning-determined residential supply $\bar{H}_i^R$.

Upper nest choice : applying the GEV choice-probability formula to (1), households compare residence-workplace pairs using the route-choice-corrected value $\tilde{V}_{ij}$, with T_i, E_j entering exactly as in the single-route model:

$$\pi_{ij} = \frac{T_i E_j \tilde{V}_{ij}^\epsilon}{\sum_{i',j' \in S^2} T_{i'} E_{j'} \tilde{V}_{i'j'}^\epsilon} = \frac{T_i E_j \tilde{V}_{ij}^\epsilon}{\Phi}, \quad (3)$$

where $\tilde{V}_{ij} = \frac{B_i(N_i, L_i) w_j}{\tau_{ij}(q_i^R)^{1-\alpha}}$ and $\tau_{ij} = \left(\sum_{r \in R_{ij}} \tau_{ijr}^{-\theta} \right)^{-1/\theta}$ is the Fréchet mean of route costs. such that the population choosing i, j pair is:

$$N_{ij} = T_i E_j \tilde{V}_{ij}^\epsilon \frac{N}{\Phi}, \quad (4)$$

Two notions of expected utility. Write $\Phi_i \equiv \sum_j T_i E_j \tilde{V}_{ij}^\epsilon$ for location i 's own contribution to $\Phi = \sum_{i'} \Phi_{i'}$, so that $N_i/N = \Phi_i/\Phi$ by (4). Two distinct expected-utility objects can be built from this, and downstream sections use both — it matters which is used where.

If a household is free to choose its residence from the *entire* metro (not just a workplace, given a residence), a standard property of Fréchet/GEV extreme-value shocks pins down its expected utility exactly. $W_i \equiv \max_j (\ln \tilde{V}_{ij} + \ln v_{ij}(\omega))$ — the best a household could do conditional on living at i — is itself Gumbel-distributed with location parameter $\ln \Phi_i$, since it is the max of independent Fréchet-shocked draws across $j \in S$; residence choice is therefore equivalent to a further logit over the $|S|$ "meta-alternatives" $i = 1, \dots, |S|$, each with systematic value $\ln \Phi_i$. Applying McFadden's classical result for extreme-value choice models to *that* reduced problem — the expected utility conditional on any single alternative being the argmax equals the same unconditional expected-max, regardless of which alternative wins — gives

$$\mathbb{E}[\text{utility} \mid \text{residence freely chosen, realized as } i] = \Gamma(1 - \frac{1}{\epsilon}) \Phi^{1/\epsilon} = \bar{U} \quad \text{for every } i,$$

the *same* value at every location. This is not a modeling assumption; it is a mechanical consequence of the compensating-shock property of Fréchet/GEV shocks: a household that ends up at a less attractive i (lower Φ_i) must have drawn an unusually favorable idiosyncratic shock to get there, and in expectation this exactly offsets i 's lower systematic value. $\bar{U}$ is therefore the right welfare object for an agent representing the *whole* population, treating where-everyone-lives as a free margin — this is exactly the role of the transportation authority and social planner, whose objectives ((22), (30)) are already $N\Gamma(1 - 1/\epsilon)\Phi^{1/\epsilon}$, built from the aggregate Φ , correctly.

A municipal government represents a different population: *incumbents*, households whose residence at i is already a given, sunk fact, not a margin the municipality is optimizing over. For such a household, only the workplace margin j remains open, and the relevant expected utility holds i exogenously fixed rather than integrating over the residence choice itself:

$$\bar{U}_i \equiv \Gamma(1 - \frac{1}{\epsilon}) \Phi_i^{1/\epsilon}, \quad (5)$$

the expected value of $\max_j (\ln \tilde{V}_{ij} + \ln v_{ij}(\omega))$ taken directly, with no further reduction over i . Unlike $\bar{U}$, $\bar{U}_i$ varies with i 's own $\bar{B}_i, \phi_i, q_i^R$ (through Φ_i) and is generically smaller than $\bar{U}$, since Φ_i is only one summand of $\Phi = \sum_{i'} \Phi_{i'}$. $\bar{U}_i$, not $\bar{U}$, is the object used in the municipal objective in §Municipal Zoning below, for exactly this reason.

Lower nest choice : conditional on choosing (i, j) , the household chooses a route $r \in R_{ij}$. Following Allen and Arkolakis (2022), we can rewrite $\tau_{ij} = \left(\sum_{r \in R_{ij}} \tau_{ijr}^{-\theta} \right)^{-1/\theta}$ such that

$$\tau = ((I - A)^{-1})^{-1/\theta}, \quad (6)$$

where A is a matrix with elements (k, l) is $d_{kl}^{-\theta}$, such that for matrix τ , the element (i, j) is τ_{ij} . Accordingly, we can derive the link intensity as follows:

$$\pi_{jk}^{kl} = \left(\frac{\tau_{ij}}{\tau_{ik} d_{kl} \tau_{lj}} \right)^\theta \quad (7)$$

Therefore I can derive the traffic on edge (k, l) as follows:

$$n_{kl} = \sum_{ij \in S^2} N_{ij} \pi_{jk}^{kl} \quad (8)$$

1.3 Firms

At each location j , competitive firms produce a freely tradable good using labor L_j and commercial floor space H_j^F :

$$Y_j = A_j L_j^\gamma (H_j^F)^{1-\gamma}, \quad (9)$$

where $A_j > 0$ is location-specific productivity and $\gamma \in (0, 1)$. Taking wages w_j and commercial rents q_j^F as given, profit maximization yields:

$$L_j = \left(\frac{\gamma A_j}{w_j} \right)^{1/(1-\gamma)} H_j^F, \quad q_j^F = (1 - \gamma) \left(\frac{\gamma}{w_j} \right)^{\gamma/(1-\gamma)} A_j^{1/(1-\gamma)}.$$

Productivity agglomeration. Local productivity is itself subject to a Marshallian externality from local employment density:

$$A_j = \bar{A}_j L_j^\zeta, \quad \bar{A}_j > 0, \quad \zeta \in (0, 1 - \gamma), \quad (10)$$

where $\bar{A}_j$ is the exogenous fundamental and ζ is the agglomeration elasticity. As with the congestion term $\phi(N_i, L_i)$ in §Municipal Zoning, individual firms are atomistic and take

A_j — hence L_j — as given when maximizing profit; the externality only appears once the labor-demand condition above is solved out in equilibrium. Substituting (10) into $L_j = (\gamma A_j / w_j)^{1/(1-\gamma)} \bar{H}_j^F$ and solving explicitly for L_j ,

$$L_j = \left[\frac{\gamma \bar{A}_j (\bar{H}_j^F)^{1-\gamma}}{w_j} \right]^{\frac{1}{1-\gamma-\zeta}}. \quad (11)$$

The restriction $\zeta < 1 - \gamma$ is required for this exponent to be finite and positive, i.e. for equilibrium local labor demand to remain strictly decreasing in w_j — a stability condition on the strength of the externality relative to the labor share, playing the same role as the $a > 1$ convexity condition already imposed on ϕ .

1.4 Municipal Zoning

Land at each location competes for residential and commercial use. Each municipality $g \in G$ chooses a zoning wedge $\{t_i\}_{i \in S^g} \geq 0$ on commercial land use at each location: procedural burden — discretionary review, permitting delay, litigation risk, compliance — drives a wedge between the residential and commercial rents the same underlying land parcel could command. Because the same underlying land parcel can be put to either use, competitive landowners allocate it to whichever use pays the higher effective return; if both uses are active in equilibrium, this competition equalizes returns at the margin:

$$t_i = \frac{q_i^F}{q_i^R} \quad (12)$$

In general $t_i > 1$ implies a pro-residential zoning, whereas $t_i < 1$ implies a pro-commercial zoning. Combined with the residential rent-clearing condition (2), the arbitrage condition (12) gives a one-to-one mapping between the zoning wedge t_i and the direct-quantity instrument $\bar{H}_i^R$:

$$t_i \frac{(1-\alpha) \sum_j N \pi_{ij} w_j}{\bar{H}_i^R} = q_i^F = (1-\gamma) \left(\frac{\gamma}{w_i} \right)^{\gamma/(1-\gamma)} A_i^{1/(1-\gamma)}. \quad (13)$$

Treating q_i^F as locally fixed (the same simplification used below and in §Transportation Authority, deferring the further GE feedback through w_i, A_i to the quantification stage) and using $\partial q_i^R / \partial \bar{H}_i^R = -q_i^R / \bar{H}_i^R$ from (2), implicit differentiation of $t_i q_i^R = q_i^F$ gives

$$\frac{d\bar{H}_i^R}{dt_i} = \frac{\bar{H}_i^R}{t_i} > 0, \quad (14)$$

so $\bar{H}_i^R(t_i)$ is strictly increasing: the map between the zoning-wedge instrument and the direct-quantity instrument is one-to-one wherever both uses are active.

The municipality sets $\{t_i\}$ to maximize the total welfare of its residents. In line with the political economy of local land use (Mast, 2024; Rollet and Weiwu, 2025), the municipal government, acting as a representative of its residents, trades off (1) lower rents and reduced worker-congestion from shrinking commercial land — both favor a higher zoning wedge — against (2) the residential crowding that comes with the resulting in-migration, and (3) the loss of local wage income to residents who both live and work at i , both of which favor a lower wedge.

$$\max_{\{t_i\}_{i \in S^g}} W^g = \sum_{i \in S^g} N_i \bar{U}_i \quad \text{s.t.} \quad (12), \quad \bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where N_i is the equilibrium population at location i and $\bar{U}_i = \Gamma(1 - \frac{1}{\epsilon}) \Phi_i^{1/\epsilon}$ — the expected utility of a household with residence i held fixed, optimizing only over workplace j (§Households, "Two notions of expected utility") — is the relevant welfare object for the municipality's *incumbent* residents, as opposed to the metro-wide $\bar{U} = \Gamma(1 - 1/\epsilon)\Phi^{1/\epsilon}$ used by the transportation authority and social planner, who represent the whole population including the free residence margin.¹ taking other municipalities' zoning wedges $\{t_{i'}\}_{i' \notin S^g}$ and transportation investment $\{I_{k\ell}\}$ as given. $\phi(N_i, L_i)$ is the congestion term that discounts the effective amenity $B_i(N_i, L_i) = \bar{B}_i/\phi(N_i, L_i)$ introduced in §Households, capturing two sources of local negative externalities: residential congestion from neighbors and the presence of workers near the residential location:

$$\frac{\partial \phi}{\partial N_i} > 0, \quad \frac{\partial \phi}{\partial L_i} > 0,$$

where N_i denotes residential population and L_i denotes employment at location i (defined above). I parametrize the disutility in Cobb-Douglas form:

$$\phi(N_i, L_i) = \eta N_i^a L_i^b, \quad \eta > 0, \quad a, b > 0, \quad (15)$$

so that $B_i(N_i, L_i) = \bar{B}_i \eta^{-1} N_i^{-a} L_i^{-b}$: an exogenous amenity level $\bar{B}_i$, marked down by local residential and workplace density. Since $\bar{H}_i^F = \bar{H}_i - \bar{H}_i^R$, the chain rule reduces the municipal first-order condition to a single comparison,

$$\frac{dW^g}{dt_i} = \left(\frac{\partial W^g}{\partial \bar{H}_i^R} - \frac{\partial W^g}{\partial \bar{H}_i^F} \right) \quad (16)$$

$$\times \frac{d\bar{H}_i^R}{dt_i} = 0, \quad (17)$$

so the optimum reduces to signing $\partial W^g / \partial \bar{H}_i^R - \partial W^g / \partial \bar{H}_i^F$, given $d\bar{H}_i^R / dt_i \neq 0$ ((14) above). The Channels paragraph below derives this comparison by implicit differentiation, treating $\partial N_i / \partial \bar{H}_i^R$ etc. as unspecified GE objects; the remainder of this subsection derives the same comparison rigorously, as a Lagrangian subject to the equilibrium conditions municipality g 's choice touches — the same "problem subject to equilibrium" treatment already used for the transportation authority (§Transportation Authority, per Open Question #17), rather than approximating the GE feedback away from the start.

The municipal problem as a Lagrangian. $N_i, L_i, \phi_i, q_i^R, w_i$ are equilibrium objects, not primitives. Following (24)'s construction, and pulling ϕ_i out as its own variable — exactly as $d_{k\ell}$ is pulled out in (24), so that N_i, L_i enter nowhere except their own defining

¹An earlier version of this objective, $N^g \cdot \bar{U}$ (aggregate expected utility times the municipality's population), was considered and rejected: since $\bar{U}$ depends only on the metro-wide Φ , which a single (small, per Open Question #11's small-open-location approximation) municipality can barely move, that objective's first-order condition has no interior solution — the municipality would always want to zone everything residential, since attracting residents is free in expected-utility terms and $d\bar{H}_i^R / \partial \bar{H}_i^R$ -type population-attraction effects are strictly positive throughout. Incumbents' residence is a given, sunk fact, not a margin they optimize over, so $\bar{U}_i$, not $\bar{U}$, is the right object.

identities and ϕ_i 's — municipality g 's problem is

$$\begin{aligned}
\mathcal{L}^g = & \sum_{i \in S^g} N_i \Gamma(1 - \frac{1}{\epsilon}) \Phi_i^{1/\epsilon} + \sum_{i \in S^g} \chi_i \left[\phi_i - \eta N_i^a L_i^b \right] + \sum_{i'j' \in S^2} \lambda_{i'j'} \left[T_{i'} E_{j'} \tilde{V}_{i'j'}^\epsilon - \pi_{i'j'} \Phi \right] \\
& + \sum_{i \in S^g} \sigma_i \left[N_i - N \sum_j \pi_{ij} \right] + \sum_{i \in S^g} \rho_i \left[L_i - N \sum_j \pi_{ji} \right] \\
& + \sum_{i \in S^g} \eta_i^q \left[q_i^R \bar{H}_i^R - (1 - \alpha) \sum_j N \pi_{ij} w_j \right] + \sum_j \eta_j^w \left[L_j(w_j) - N \sum_i \pi_{ij} \right] \\
& + \sum_{i \in S^g} \nu_i \left[\bar{H}_i - \bar{H}_i^R - \bar{H}_i^F \right], \tag{18}
\end{aligned}$$

where $\Phi_i \equiv \bar{B}_i^\epsilon \phi_i^{-\epsilon} \sum_j T_i E_j w_j^\epsilon \tau_{ij}^{-\epsilon} (q_i^R)^{-\epsilon(1-\alpha)}$ as before, and $\tilde{V}_{ij} = \bar{B}_i w_j / (\phi_i \tau_{ij} (q_i^R)^{1-\alpha})$. $\lambda_{i'j'}$ (household optimality, all OD pairs — a zoning change at $i \in S^g$ moves q_i^R and ϕ_i , hence $\tilde{V}_{ij}$, hence potentially every $\pi_{i'j'}$ through Φ), η_i^q (rent clearing), and η_j^w (labor clearing, $L_j(w_j) \equiv (\gamma A_j / w_j)^{1/(1-\gamma)} \bar{H}_j^F$ under (11)) reuse the same multipliers already carrying these meanings in (24); ν_i (land identity) reuses the one already used in (30). Three multipliers are new: χ_i (the pulled-out congestion identity), and σ_i, ρ_i on the population and employment identities $N_i = N \sum_j \pi_{ij}$, $L_i = N \sum_j \pi_{ji}$. These are needed because $\bar{U}_i$ enters the municipal objective multiplying N_i *outside* Φ_i , in addition to N_i, L_i determining ϕ_i *inside* Φ_i — a role N_i, L_i never play in the transportation authority's or planner's objective, which depends on the aggregate Φ only. No separate multiplier is needed for the arbitrage condition (12): $\bar{H}_i^R$ (not t_i) is the genuine Lagrangian choice variable, and t_i^* is recovered afterward exactly as in Definition 1.

N_i, L_i : clean first-order conditions. Having pulled ϕ_i out, N_i and L_i appear only in the objective's front factor, the χ_i bracket, and their own identity bracket:

$$\frac{\partial \mathcal{L}^g}{\partial N_i} = 0 : \quad \sigma_i = \chi_i \frac{a \phi_i}{N_i} - \bar{U}_i, \quad \frac{\partial \mathcal{L}^g}{\partial L_i} = 0 : \quad \rho_i = \chi_i \frac{b \phi_i}{L_i}. \tag{19}$$

$\bar{H}_i^R, \bar{H}_i^F$ each appear only in their own market-clearing bracket and the shared land-identity bracket — no $\lambda_{i'j'}$ appearance, since neither enters $\tilde{V}_{ij}$ directly — giving the clean pair

$$\frac{\partial \mathcal{L}^g}{\partial \bar{H}_i^R} = 0 : \quad \nu_i = \eta_i^q q_i^R, \quad \frac{\partial \mathcal{L}^g}{\partial \bar{H}_i^F} = 0 : \quad \nu_i = \eta_i^w \left. \frac{\partial L_i}{\partial \bar{H}_i^F} \right|_{w_i \text{ fixed}} = \eta_i^w \frac{(1 - \gamma) L_i}{(1 - \gamma - \zeta) \bar{H}_i^F}, \tag{20}$$

the last equality exact from (11). Combining, $\eta_i^q q_i^R = \eta_i^w (1 - \gamma) L_i / [(1 - \gamma - \zeta) \bar{H}_i^F]$ — the rigorous, shadow-price version of "signing $\partial W^g / \partial \bar{H}_i^R - \partial W^g / \partial \bar{H}_i^F$," replacing the informal statement in (16).

Closing the system: ϕ_i, q_i^R, w_i . $\chi_i, \eta_i^q, \eta_j^w$ are not yet closed: ϕ_i, q_i^R, w_i each enter $\tilde{V}_{ij}$ directly, so each appears not only in the objective and its own market-clearing bracket but inside *every* $\lambda_{i'j'}$ bracket touching it — exactly the situation τ_{ij} and $d_{k\ell}$ are in for the transportation authority (§Transportation Authority, "Closing the system"), resolved there by defining the λ -weighted market-access object $\Lambda_{k\ell}$. The same construction applies here: differentiating $\mathcal{L}^g$ in ϕ_i gives $\chi_i = (N_i \bar{U}_i + \epsilon \Lambda_i^\phi) / \phi_i$ with $\Lambda_i^\phi \equiv \sum_j \lambda_{ij} T_i E_j \tilde{V}_{ij}^\epsilon$; the analogous q_i^R - and w_i -conditions define Λ_i^q, Λ_i^w the same way, with $\Lambda_i^w \equiv \sum_{i' \in S} \lambda_{i'i} T_{i'} E_i \tilde{V}_{i'i}^\epsilon$

running over the *whole* metro — every location whose residents might commute to work-place i — not just S^g . As with $\Lambda_{k\ell}$, these are not expanded symbol-by-symbol here: they solve their own linear system jointly with every $\lambda_{i'j'}$, the same size and type of object as §Transportation Authority's, deferred to quantification.

Channels. Only the i -th term of $W^g = \sum_{i' \in S^g} N_{i'} \bar{U}_{i'}$ depends on $\bar{H}_i^R, \bar{H}_i^F$ to leading order, holding other locations' land use and the metro-wide index Φ fixed. With the congestion term now in the amenity, $\Phi_i = \bar{B}_i^\epsilon \phi_i^{-\epsilon} \sum_j T_j E_j w_j^\epsilon \tau_{ij}^{-\epsilon} (q_i^R)^{-\epsilon(1-\alpha)}$, and the partials of $\bar{U}_i = \Gamma(1 - 1/\epsilon) \Phi_i^{1/\epsilon}$ are exact and closed-form:

$$\frac{\partial \bar{U}_i}{\partial q_i^R} = -\frac{(1-\alpha)\bar{U}_i}{q_i^R}, \quad \frac{\partial \bar{U}_i}{\partial \phi_i} = -\frac{\bar{U}_i}{\phi_i}, \quad \frac{\partial \bar{U}_i}{\partial w_i} = \frac{\pi_{ii|i} \bar{U}_i}{w_i} > 0,$$

where $\pi_{ii|i}$ is the conditional probability that a resident of i also works at i (the local live-work share): the first equality is exact since q_i^R enters Φ_i as a common factor across every j ; the second is exact because ϕ_i enters Φ_i as the single power $\phi_i^{-\epsilon}$, common across every j — no first-order approximation is needed, unlike the earlier income-side formulation where ϕ_i was netted against w_j ; the third holds because only the $j = i$ term of Φ_i depends on w_i , so the wage channel is scaled down by how many residents actually capture it, and is likewise simpler than before since w_i no longer nets against ϕ_i .

Substituting these, together with $\phi_i = \eta N_i^a L_i^b$, into $\partial(N_i \bar{U}_i)/\partial \bar{H}_i^R - \partial(N_i \bar{U}_i)/\partial \bar{H}_i^F$ gives the explicit optimality condition

$$\begin{aligned} & \underbrace{\bar{U}_i \frac{\partial N_i}{\partial \bar{H}_i^R}}_{(1) \text{ in-mover } \geq 0} + \underbrace{N_i \frac{\partial \bar{U}_i}{\partial q_i^R} \frac{\partial q_i^R}{\partial \bar{H}_i^R}}_{(2) \text{ rent } \geq 0} + \underbrace{N_i \frac{\partial \bar{U}_i}{\partial \phi_i} \eta a N_i^{a-1} L_i^b \frac{\partial N_i}{\partial \bar{H}_i^R}}_{(3) \text{ residential congestion } \leq 0} \\ & - \underbrace{N_i \frac{\partial \bar{U}_i}{\partial \phi_i} \eta b N_i^a L_i^{b-1} \frac{\partial L_i}{\partial \bar{H}_i^F}}_{(4) \text{ worker-congestion relief } \geq 0} - \underbrace{N_i \frac{\partial \bar{U}_i}{\partial w_i} \frac{\partial w_i}{\partial \bar{H}_i^F}}_{(5) \text{ local-wage loss } \leq 0} = 0. \end{aligned} \quad (21)$$

Terms (1)–(5) reproduce the trade-off: attracting residents and lowering rents (1)–(2), against the residential crowding those in-movers bring (3), net of the worker-congestion relief from shrinking commercial land (4), shrinking $\bar{H}_i^F$ contracts local labor demand along an upward-sloping labor supply curve — household workplace choice responds to w_i with finite elasticity ϵ (5). The municipality thus weighs the value of local jobs to its own live-work residents against the congestion those same jobs create. With the productivity agglomeration externality (10), term (5) is larger in magnitude than in a model with exogenous A_i : shrinking $\bar{H}_i^F$ lowers L_i along the labor-demand curve (11) *and* lowers local productivity $A_i = \bar{A}_i \zeta_i^L$ directly, so $\partial w_i / \partial \bar{H}_i^F$ now compounds a pure quantity effect with a genuine productivity loss — the local-wage channel through which zoning restriction is costly to residents is strictly stronger once agglomeration is present. Given the monotonicity of $H_i^R(t_i)$ ((14)), the municipal first-order condition (16) on t_i is equivalent to the first-order condition on $\bar{H}_i^R$.

1.5 Transportation Authority

The metropolitan transportation authority chooses investment $\{I_{k\ell}\}_{(k,\ell) \in E}$ to maximize aggregate household welfare, taking the full zoning vector $\{\bar{H}_i^R, \bar{H}_i^F\}$ and the route

assignment $\{r_{ij}\}$ as given. Following Bordeu (2026)'s formalization of her local government's problem (her §2.1.4), the maximization is stated with the equilibrium conditions of the rest of the model entered directly as constraints, alongside the budget constraint, rather than solved for separately beforehand:

$$\max_{\{I_{k\ell}\}_{(k,\ell) \in E}} N \cdot \Gamma\left(1 - \frac{1}{\epsilon}\right) \Phi^{1/\epsilon}, \quad \Phi \equiv \sum_{i'j'} T_{i'} E_{j'} \tilde{V}_{i'j'}^\epsilon \quad (22)$$

subject to (i) household optimality: $\{\pi_{ij}\}$ given by (3)–(4);

(ii) commuting-cost index: τ_{ij} given by (6);

(iii) edge-level commuting cost: $d_{k\ell} = \exp(\kappa \text{time}_{k\ell}(I_{k\ell}, n_{k\ell}))$ (§Transportation);

(iv) edge traffic: $n_{k\ell} = \sum_{ij \in S^2} N \pi_{ij} \pi_{ij}^{k\ell}$ (8);

(v) labor market clearing: $N \sum_i \pi_{ij} = L_j(w_j) \equiv (\gamma A_j / w_j)^{1/(1-\gamma)} \bar{H}_j^F$ (§Firms);

(vi) residential rent clearing: q_i^R given by (2);

(vii) budget: $\sum_{(k,\ell) \in E} c_{k\ell} I_{k\ell} \leq K, \quad I_{k\ell} \geq 0,$

where $c_{k\ell}$ is the per-unit cost of investment on edge (k, ℓ) and K is the total infrastructure budget. Constraints (ii) and (iii) split what was previously described only in prose as "route choice": (ii) is the bilateral commuting-cost index aggregating route costs within an OD pair (Bordeu 2026's analogous constraint, her eq. 12), and (iii) is the edge-level cost technology itself (her eq. 10) — kept as separate constraints, following her construction exactly, rather than substituted directly, because $I_{k\ell}$ enters the whole system only through (iii). Constraint (v) folds *firm optimality* directly into *labor-market clearing*: the firm's labor demand $L_j(w_j)$ is a simple closed-form function of w_j (§Firms), not an implicit/coupled object, so it is substituted in rather than given its own multiplier — the same combination Bordeu uses for her analogous wage/labor-clearing condition. Constraints (i)–(vi) are solved jointly, not sequentially: (iii) feeds back into (ii) and hence (i), (iv) feeds back into (iii) through $n_{k\ell}$, while (v)–(vi) simultaneously pin down w_j and q_i^R , which enter $\tilde{V}_{ij}$ inside Φ . This system defines $\{\pi_{ij}(I), \tau_{ij}(I), d_{k\ell}(I), n_{k\ell}(I), w_j(I), q_i^R(I)\}$ as equilibrium objects rather than free parameters, and hence also $N_i(I) = N \sum_j \pi_{ij}(I)$ and $L_i(I) = N \sum_j \pi_{ji}(I)$; Φ in the objective is the resulting reduced-form value, a function of I through this whole system, not only through τ_{ij} directly.

Direct edge elasticity. From $d_{k\ell} = \exp(\kappa \text{time}_{k\ell})$ and $\text{time}_{k\ell} = \delta_{k\ell}^0 + \delta_{k\ell}^1 n_{k\ell}^\rho / I_{k\ell}^\beta$ (§Transportation),

$$\frac{\partial \ln d_{k\ell}}{\partial I_{k\ell}} = -\frac{\kappa \beta \delta_{k\ell}^1 n_{k\ell}^\rho}{I_{k\ell}^{\beta+1}} \quad (\text{holding } n_{k\ell} \text{ fixed}), \quad \frac{\partial \ln d_{k\ell}}{\partial n_{k\ell}} = \frac{\kappa \rho \delta_{k\ell}^1 n_{k\ell}^{\rho-1}}{I_{k\ell}^\beta}. \quad (23)$$

Route-choice bookkeeping. Differentiating $\tau_{ij}^{-\theta} = \sum_{r \in R_{ij}} \tau_{ijr}^{-\theta}$ and using $\ln \tau_{ijr} = \sum_{(k,\ell) \in r} \ln d_{k\ell}$,

$$\frac{\partial \ln \tau_{ij}}{\partial d_{k\ell}} = \sum_{r \in R_{ij}} \pi_{r|ij} \frac{\partial \ln \tau_{ijr}}{\partial d_{k\ell}} = \frac{1}{d_{k\ell}} \sum_{r \ni (k,\ell)} \pi_{r|ij} = \frac{\pi_{ij}^{k\ell}}{d_{k\ell}} \quad (\text{holding all other } d\text{'s fixed}),$$

where $\pi_{r|ij} = (\tau_{ij} / \tau_{ijr})^\theta$ is the conditional route-choice probability and $\pi_{ij}^{k\ell} = \sum_{r \ni (k,\ell)} \pi_{r|ij}$ is exactly the edge-intensity object of (7): the probability that an (i, j) -commuter's chosen route uses edge (k, ℓ) . This is the object that links τ_{ij} to $d_{k\ell}$ below; it does not yet involve $I_{k\ell}$ or $n_{k\ell}$ directly, since those enter only through $d_{k\ell}$ itself, constraint (iii).

Formalizing "subject to equilibrium": the Lagrangian. $\tau_{ij}, d_{kl}, w_j, q_i^R, N_i, L_i$ (and, through them, $\tilde{V}_{ij}$) are equilibrium objects, not primitives — they respond to I_{kl} through the constraint system above. Following Bordeu (2026)'s treatment of her local government's problem (her §2.1.4 and Appendix A.3, which formalizes exactly this "planner subject to equilibrium" structure by entering every equilibrium condition into the Lagrangian with its own multiplier, rather than approximating their effect away), I write the authority's problem as a single Lagrangian over constraints (i)–(vi) of (22) — six multipliers, one per constraint, matching Bordeu's own six ($\lambda_{ij}, \eta_{Ri}, \eta_{Fj}, \gamma_{ij}, \epsilon_{kl}, \phi_{kl}$ in her notation) rather than substituting any of them away:

$$\begin{aligned} \mathcal{L} = & N\Gamma(1 - \tfrac{1}{\epsilon}) \Phi^{1/\epsilon} - \mu \left(\sum_{kl} c_{kl} I_{kl} - K \right) + \sum_{ij} \lambda_{ij} \left[T_i E_j \tilde{V}_{ij}^\epsilon - \pi_{ij} \Phi \right] \\ & + \sum_{ij} \gamma_{ij} \left[\tau_{ij} - ((I - A)^{-1})^{-1/\theta} \right] + \sum_{kl} \psi_{kl} \left[d_{kl} - \exp(\kappa \text{time}_{kl}(I_{kl}, n_{kl})) \right] \\ & + \sum_{kl} \varphi_{kl} \left[n_{kl} - \sum_{ij} N \pi_{ij} \pi_{ij}^{kl} \right] + \sum_i \eta_i^q \left[q_i^R \bar{H}_i^R - (1 - \alpha) \sum_j N \pi_{ij} w_j \right] \\ & + \sum_j \eta_j^w \left[L_j(w_j) - N \sum_i \pi_{ij} \right], \end{aligned} \quad (24)$$

where $\lambda_{ij}, \gamma_{ij}, \psi_{kl}, \varphi_{kl}, \eta_i^q, \eta_j^w$ are the multipliers on constraints (i)–(vi) respectively. λ_{ij} is the metro-wide marginal social value of an additional household choosing the (i, j) commute pattern; γ_{ij} is the shadow value of the commuting-cost index τ_{ij} ; ψ_{kl} is the shadow value of the edge-level cost technology d_{kl} itself; φ_{kl} is the shadow value of an additional unit of traffic on edge (k, ℓ) .

First-order conditions for I_{kl} and n_{kl} . With d_{kl} pulled out as its own variable, I_{kl} and n_{kl} each appear in $\mathcal{L}$ *only* inside their own constraint-(iii) bracket — I_{kl} nowhere else at all, n_{kl} also in its own constraint-(iv) bracket. This is strictly simpler than differentiating through every OD pair routed through the edge directly. Differentiating,

$$\mu c_{kl} = -\psi_{kl} \frac{\partial d_{kl}}{\partial I_{kl}} = -\psi_{kl} d_{kl} \frac{\partial \ln d_{kl}}{\partial I_{kl}}, \quad (25)$$

$$\varphi_{kl} = \psi_{kl} \frac{\partial d_{kl}}{\partial n_{kl}} = \psi_{kl} d_{kl} \frac{\partial \ln d_{kl}}{\partial n_{kl}}. \quad (26)$$

Dividing (25) by (26) already eliminates $\psi_{kl} d_{kl}$ and gives the ratio condition below, *before* ψ_{kl} itself is pinned down — the level of I_{kl} relative to φ_{kl} does not require solving the rest of the multiplier system.

Closing the system: from λ_{ij} to ψ_{kl} . ψ_{kl} is itself determined by $\mathcal{L}$'s first-order condition in d_{kl} , which appears in its own bracket and in every γ_{ij} bracket for OD pairs routed through (k, ℓ) (via $\partial \ln \tau_{ij} / \partial d_{kl} = \pi_{ij}^{kl} / d_{kl}$, above); γ_{ij} in turn is determined by the FOC in τ_{ij} , which appears in its own bracket and in the λ_{ij} bracket (via $\tilde{V}_{ij} \propto \tau_{ij}^{-1}$). Tracing this chain through — the same linear system flagged in §Transportation Authority's earlier discussion, not expanded symbol-by-symbol here any more than Bordeu expands hers in her main text — reproduces exactly

$$\psi_{kl} d_{kl} = \epsilon \sum_{(i,j):(k\ell) \in r_{ij}} \lambda_{ij} T_i E_j \tilde{V}_{ij}^\epsilon \pi_{ij}^{kl} \equiv \epsilon \Lambda_{kl},$$

so that (25)–(26) read $\mu c_{k\ell} = -\epsilon(\partial \ln d_{k\ell}/\partial I_{k\ell})\Lambda_{k\ell}$ and $\varphi_{k\ell} = \epsilon(\partial \ln d_{k\ell}/\partial n_{k\ell})\Lambda_{k\ell}$ — identical in form to a direct-substitution derivation, but now every multiplier in the chain $(\lambda_{ij} \rightarrow \gamma_{ij} \rightarrow \psi_{k\ell} \rightarrow \varphi_{k\ell})$ is a named, separately-constrained object rather than collapsed into one step. I call $\Lambda_{k\ell}$ edge (k, ℓ) 's *market access*: the traffic-intensity-weighted sum of accessibility gains it delivers across every OD pair routed through it, each pair weighted by its own shadow value λ_{ij} rather than treated symmetrically. All six multiplier families — $\lambda_{ij}, \gamma_{ij}, \psi_{k\ell}, \varphi_{k\ell}, \eta_i^q, \eta_j^w$ — solve their own first-order conditions jointly, a linear system coupled because a change in any π_{ij} moves N_i, L_i (hence the congestion amenity B_i , hence every $\tilde{V}_{ij'}$), every route's cost index, every edge's traffic, every location's rent, and every workplace's wage simultaneously. I do not expand this system symbol-by-symbol here — Bordeu does not either, in her main text — beyond noting it is the same size and type of object as her " $2N + E$ "-plus-route/edge-cost system (her Appendix A.3).

The optimal-investment condition. Dividing (25) by (26) eliminates $\Lambda_{k\ell}$ and gives, using the model's actual $d_{k\ell}$,

$$\mu c_{k\ell} = -\varphi_{k\ell} \frac{\partial d_{k\ell}/\partial I_{k\ell}}{\partial d_{k\ell}/\partial n_{k\ell}} = \varphi_{k\ell} \frac{\beta}{\rho} \frac{n_{k\ell}}{I_{k\ell}}, \quad (27)$$

structurally identical to Bordeu (2026)'s eq. (18). Substituting (26) into (27) and solving for $I_{k\ell}$,

$$I_{k\ell} = \left(\frac{\epsilon \kappa \beta}{\mu} \right)^{1/(\beta+1)} W_{k\ell}^{1/(\beta+1)}, \quad W_{k\ell} \equiv \frac{\delta_{k\ell}^1 n_{k\ell}^\rho \Lambda_{k\ell}}{c_{k\ell}}. \quad (28)$$

Substituting into the budget constraint $\sum_{k'\ell'} c_{k'\ell'} I_{k'\ell'} = K$ (binding at the optimum, since the marginal benefit in (25) is unbounded as $I_{k\ell} \rightarrow 0^+$ whenever $W_{k\ell} > 0$) pins down $(\epsilon \kappa \beta / \mu)^{1/(\beta+1)} = K / \sum_{k'\ell'} c_{k'\ell'} W_{k'\ell'}^{1/(\beta+1)}$, eliminating μ and giving the closed-form optimal investment rule

$$I_{k\ell}^* = K \cdot \frac{W_{k\ell}^{1/(\beta+1)}}{\sum_{(k'\ell') \in E} c_{k'\ell'} W_{k'\ell'}^{1/(\beta+1)}}, \quad W_{k\ell} \equiv \frac{\delta_{k\ell}^1 n_{k\ell}^\rho \Lambda_{k\ell}}{c_{k\ell}}. \quad (29)$$

This is a budget-share rule: edge (k, ℓ) 's share of the infrastructure budget rises with its own λ -weighted market access $\Lambda_{k\ell}$ and its equilibrium traffic $n_{k\ell}$ (through ρ : busier edges get more investment, a "fix the bottleneck" force), and falls with its own cost $c_{k\ell}$. It automatically handles the corner case: if $W_{k\ell} = 0$ (no OD pair routed through (k, ℓ) , or zero equilibrium traffic under $\rho > 0$), then $I_{k\ell}^* = 0$, no separate complementary-slackness case needed.

1.6 Equilibrium

Definition 1 (Decentralized Equilibrium). A decentralized equilibrium is a profile

$$\left(\{I_{k\ell}^*\}_{(k,\ell) \in E}, \{t_i^*\}_{i \in S}, \{N\pi_{ij}^*\}_{i,j \in S}, \{n_{k\ell}^*\}_{(k,\ell) \in E}, \{q_i^{R*}, q_i^{F*}, w_i^*\}_{i \in S} \right)$$

such that:

1. **Household optimality.** Commuting flows $\{N\pi_{ij}^*\}$ are given by (3)–(4) at prices (q_i^{R*}, w_i^*) and investment I^* , with route choice by (6) and route-based commuting costs $\{\tau_{ij}^*\}$.
2. **Edge traffic.** $\{n_{k\ell}^*\}$ satisfies (8) given $\{\pi_{ij}^*\}$ and route assignment $\{r_{ij}\}$, jointly with (1) since $n_{k\ell}^*$ feeds back into $d_{k\ell}(I_{k\ell}^*, n_{k\ell}^*)$ and hence τ_{ij}^* .
3. **Firm optimality.** Wages w_i^* and commercial rents q_i^{F*} satisfy the firm's first-order conditions at $H_i^F = \bar{H}_i^{F*}$, with labor demand L_i^* solving (11) under the productivity externality (10).
4. **Market clearing.** Residential floor space clears at q_i^{R*} given by (2), commercial floor space is fixed at $H_i^F = \bar{H}_i^{F*}$, and the labor market clears at $L_i^* = N \sum_j \pi_{ji}^*$.
5. **Municipal Nash equilibrium.** Each municipality g sets the zoning wedge $\{t_i^*\}_{i \in S^g}$ — equivalently, induces $\{\bar{H}_i^{R*}, \bar{H}_i^{F*}\}_{i \in S^g}$ via (14) — to satisfy the first-order condition (16), given $\{\bar{H}_{i'}^{R*}, \bar{H}_{i'}^{F*}\}_{i' \notin S^g}$ and I^* ; no municipality has a profitable deviation.
6. **Transportation authority optimality.** $\{I_{k\ell}^*\}$ solves (22), given $\{\bar{H}_i^{R*}, \bar{H}_i^{F*}\}$: equivalently, $I_{k\ell}^*$ satisfies (27) for every edge $(k, \ell) \in E$, where the multipliers $(\lambda_{ij}^*, \gamma_{ij}^*, \psi_{k\ell}^*, \varphi_{k\ell}^*, \eta_i^{q*}, \eta_j^{w*})$ solve their own first-order conditions from (24) jointly with (1)–(4) at the equilibrium profile.

Conditions (v) and (vi) jointly constitute the simultaneous Nash equilibrium between municipalities and the transportation authority: each set of agents takes the other's choices as given and best-responds. The equilibrium zoning wedge $t_i^* \equiv q_i^{F*}/q_i^{R*}$ is computed from the equilibrium quantities in this profile, not solved for as a separate object.

1.7 Scale of the Zoning Decision

Widening the zoning decision from a single municipality to the whole metro can change two things at once: the *scope* of locations one zoning-setter controls (S^g versus S), and *whose* welfare that setter maximizes. For a genuinely metropolitan authority — one no longer representing a fixed set of incumbents at a handful of locations, but the region as a whole — the natural objective is the same population-wide $\bar{U}$ the transportation authority and social planner already use (§Households, "Two notions of expected utility"), not the incumbent-style $\bar{U}_i$ that is specifically motivated by a municipality representing residents whose residence is held fixed. This subsection's benchmark therefore changes scope *and* objective together, and shows the two changes combine cleanly: the resulting zoning condition turns out to be exactly the social planner's.

The Metropolitan Zoning Authority. Consider replacing the $|G|$ municipalities with a single zoning authority that chooses $\{t_i\}_{i \in S}$ — the zoning wedge at *every* location — to maximize the same aggregate welfare as the transportation authority and social planner,

$$\max_{\{t_i\}_{i \in S}} N\Gamma(1 - \frac{1}{\epsilon}) \Phi^{1/\epsilon} \quad \text{s.t.} \quad (12), \quad \bar{H}_i^R + \bar{H}_i^F = \bar{H}_i \quad \forall i \in S,$$

still taking the transportation authority's investment $\{I_{k\ell}\}$ as given, with one zoning agent replacing $|G|$.

FOC. The metropolitan authority's Lagrangian is §Social Planner's — $\chi_i, \sigma_i, \rho_i, \lambda_{i'j'}, \eta_i^q, \eta_j^w, \nu_i$ over the aggregate objective, extending (24) with the land identity exactly as (30) does — *minus* the edge/route constraints (ii)–(iv) and the budget multiplier μ , since the authority does not choose $I_{k\ell}$; it takes the transportation authority's current investment as given. But $\bar{H}_i^R$ never touched those constraints in the social planner's version either — it enters the Lagrangian only through the rent-clearing bracket (η_i^q) and the land-identity bracket (ν_i), exactly as in §Municipal Zoning and §Social Planner. Removing the pieces the metropolitan authority doesn't control changes nothing about the $\bar{H}_i^R$ -first-order condition, so the authority's zoning condition is exactly (33).

Result. With the objective and scope now matching the social planner's, §Social Planner's "why investment does not add a term" argument extends directly: it only used that $I(t)$ is the transportation authority's own optimal response to whatever zoning is currently in place, never that zoning and investment were chosen jointly. So it applies equally whether the aggregate-objective zoning-setter is in Nash with the transportation authority (this subsection) or jointly optimizing investment with it (§Social Planner) — **agent identity is welfare-irrelevant for the zoning margin**, given that the objective and scope match. This reinstates the original (2026-07-21) conclusion, now because the objectives were verified to match rather than assumed to. What remains a genuine, economically meaningful comparison is municipal (incumbent-style $\sum_{i \in S^g} N_i \bar{U}_i$, local) against the metropolitan authority or planner (population-wide $\bar{U}$, regional) — a comparison that changes scope *and* objective together, which is exactly what §Welfare Comparison evaluates (decentralized equilibrium versus social planner).

1.8 Social Planner

The decentralized equilibrium splits control of the metro's two instruments across two kinds of agents, each at its own scale: zoning is chosen municipality-by-municipality, each maximizing only over its own S^g ; investment is chosen by a metropolitan authority, but taking zoning as given. The natural benchmark isolates both distortions at once — a single metropolitan government choosing *both* instruments, for the *whole* metro, jointly. Following Bordeu (2026)'s comparison of her decentralized and "Centralized Planner (Metropolitan)" cases, and using exactly the Lagrangian apparatus of §Transportation Authority, the planner's problem is

$$\begin{aligned} \max_{\{t_i\}_{i \in S}, \{I_{k\ell}\}_{(k,\ell) \in E}} \quad & N \cdot \Gamma(1 - \tfrac{1}{\epsilon}) \Phi^{1/\epsilon} \\ \text{subject to} \quad & \text{(i)–(vii) of (22), and} \\ & \text{(viii) land identity: } \bar{H}_i^R + \bar{H}_i^F = \bar{H}_i \quad \forall i \in S. \end{aligned} \tag{30}$$

The only difference from (22) is that $\{\bar{H}_i^R, \bar{H}_i^F\}_{i \in S}$ — equivalently $\{t_i\}_{i \in S}$, via (12) — are now genuine choice variables for every location, not parameters the constraint system takes as given; correspondingly, (24) gains one term, $\sum_i \nu_i [\bar{H}_i - \bar{H}_i^R - \bar{H}_i^F]$, with multiplier ν_i on the land identity at i .

Investment. $I_{k\ell}$ enters the planner's Lagrangian through exactly the same terms as in (24) — the objective and constraints (i)–(vii) are unchanged — so the FOC is unchanged

in form:

$$I_{k\ell}^{SP} = K \cdot \frac{W_{k\ell}^{1/(\beta+1)}}{\sum_{(k'\ell') \in E} c_{k'\ell'} W_{k'\ell'}^{1/(\beta+1)}}, \quad W_{k\ell} \equiv \frac{\delta_{k\ell}^1 n_{k\ell}^\rho \Lambda_{k\ell}}{c_{k\ell}}, \quad (31)$$

identical to (29). The planner's investment rule is the *same formula*, evaluated at the jointly optimal zoning vector $\{\bar{H}_i^{R,SP}\}$ instead of the Nash-equilibrium one $\{\bar{H}_i^{R,DE}\}$ — the entire wedge between $I_{k\ell}^{SP}$ and $I_{k\ell}^{DE}$ operates through how zoning shifts $n_{k\ell}$ and $\Lambda_{k\ell}$, not through any change in the investment rule itself.

Zoning. $\bar{H}_i^R$ enters the planner's Lagrangian only through rent clearing and the land identity, exactly as in §Municipal Zoning — so (24), extended with ν_i , needs the identical χ_i, σ_i, ρ_i additions to correctly track N_i, L_i, ϕ_i ((18)'s construction, now over the whole S , with the planner's own objective $N\Gamma(1 - 1/\epsilon)\Phi^{1/\epsilon}$ in place of $\sum N_i \bar{U}_i$). That objective difference changes two things relative to §Municipal Zoning's clean conditions (19):

- N_i has no front-factor role in $\Phi^{1/\epsilon}$ — population is entirely *derived* from Φ ($N_i = N\Phi_i/\Phi$), never counted separately the way $\sum N_i \bar{U}_i$ counts it. So $\partial \mathcal{L}^{SP}/\partial N_i = 0$ gives $\sigma_i = \chi_i a \phi_i / N_i$, with *no* $-\bar{U}_i$ term: the planner's zoning margin has no analogue of channel (1), the in-mover term.
- Wherever $\bar{U}_i$ appeared in §Municipal Zoning's closing conditions, the planner's version has the *aggregate* $\bar{U} \equiv \Gamma(1 - 1/\epsilon)\Phi^{1/\epsilon}$ instead, e.g.

$$\eta_i^q = \frac{(1 - \alpha)[N_i \bar{U} + \epsilon \Lambda_i^q]}{q_i^R \bar{H}_i^R}, \quad (32)$$

using the same $\Lambda_i^q \equiv \sum_j \lambda_{ij} T_i E_j \tilde{V}_{ij}^\epsilon$ as §Municipal Zoning — a change in i 's rent is weighed by how it moves the *whole-metro* $\bar{U}$, not a location-specific $\bar{U}_i$.

Combined with the analogous η_i^w (closing through Λ_i^w), the planner's zoning condition is

$$\frac{(1 - \alpha)[N_i \bar{U} + \epsilon \Lambda_i^q]}{\bar{H}_i^R} = \eta_i^w \frac{(1 - \gamma)L_i}{(1 - \gamma - \zeta)\bar{H}_i^F}. \quad (33)$$

This is *not* (21)'s five channels plus a spillover term — it is a structurally different condition: missing channel (1), using $\bar{U}$ everywhere $\bar{U}_i$ appeared, and with $\Lambda_i^\phi, \Lambda_i^q, \Lambda_i^w$ genuinely live rather than collapsible, for the reason given in §Scale of the Zoning Decision: the planner spans the whole S , so no location it rezones is negligible relative to the Φ that rezoning moves.

An earlier version of this result (2026-07-21, restated 2026-07-23) claimed (33) reduces to (21) plus one clean additive "cross-jurisdiction spillover" term, matching a metropolitan zoning authority that was, at the time, given the municipal-style objective $\sum_{i \in S} N_i \bar{U}_i$ — conflating two different objectives, since the planner here uses the population-wide $\bar{U}$, not the incumbent-style $\bar{U}_i$ (§Households, "Two notions of expected utility"). With the metropolitan zoning authority's objective corrected to match the planner's (§Scale of the Zoning Decision now uses the same aggregate $\bar{U}$), the two conditions coincide exactly: §Scale of the Zoning Decision's authority solves (33) too, since removing the edge/route constraints and the budget multiplier it doesn't have changes nothing about the $\bar{H}_i^R$ -first-order condition. **Agent identity is welfare-irrelevant for the zoning margin** — but only once the objective and scope genuinely match, which is now verified rather than assumed.

Why investment does not add a term. $\bar{H}_i^R$ and $I_{k\ell}$ are chosen *jointly* at the planner's optimum, each with its own first-order condition ($\partial \mathcal{L}^{SP} / \partial \bar{H}_i^R = 0$, giving (33), and, from (25), $\partial \mathcal{L}^{SP} / \partial I_{k\ell} = 0$ for every edge). Because $\partial \mathcal{L}^{SP} / \partial I_{k\ell} = 0$ holds at the optimum for every (k, ℓ) , any chain-rule term of the form $(\partial \mathcal{L}^{SP} / \partial I_{k\ell})(\partial I_{k\ell}^{SP} / \partial \bar{H}_i^R)$ is identically zero — there is no separate "feedback through I " term to add to (33), regardless of what the rest of that condition looks like. The same conclusion follows without reference to the Lagrangian: define $\tilde{W}(t) \equiv W(t, I^{SP}(t))$; by the envelope theorem, $d\tilde{W}/dt_i = \partial W / \partial t_i + \sum_{k\ell} (\partial W / \partial I_{k\ell})(dI_{k\ell}^{SP} / dt_i)$, and $\partial W / \partial I_{k\ell} = \mu c_{k\ell}$ at the optimum ((25)) while $\sum_{k\ell} c_{k\ell} I_{k\ell}^{SP}(t) = K$ binds for every t (fixed K), so $\sum_{k\ell} c_{k\ell} dI_{k\ell}^{SP} / dt_i = dK / dt_i = 0$ and the second sum vanishes. Both checks confirm (33) needs no investment-feedback term — the correction needed relative to the earlier (2026-07-21/23) version is entirely in the zoning condition's own leading terms (missing channel (1), $\bar{U}$ in place of $\bar{U}_i$, live Λ_i 's), not in whether investment contributes an extra piece. Neither argument actually used that $I_{k\ell}$ was chosen *jointly* with $\bar{H}_i^R$ — only that $I(t)$ is the transportation authority's own optimal response to whatever zoning is current — so both extend unchanged to a zoning-setter in Nash with the transportation authority rather than choosing I itself, which is exactly §Scale of the Zoning Decision's Metropolitan Zoning Authority.

1.9 Welfare Comparison

This subsection states and proves the welfare gap between the decentralized equilibrium (DE) and the social planner (SP), in two parts: a global ranking that requires no derivatives, and a first-order characterization of what drives it.

Proposition 1 (Planner dominance). Let $W^{DE} \equiv N\Gamma(1 - 1/\epsilon)(\Phi^{DE})^{1/\epsilon}$ and $W^{SP} \equiv N\Gamma(1 - 1/\epsilon)(\Phi^{SP})^{1/\epsilon}$ denote aggregate household welfare at the decentralized equilibrium profile (t^{DE}, I^{DE}) (Definition 1) and at the social planner's solution (t^{SP}, I^{SP}) of (30), respectively. Then

$$W^{SP} \geq W^{DE}.$$

Proof. The decentralized profile (t^{DE}, I^{DE}) satisfies constraints (i)–(vi) of (22) (Definition 1, conditions 1–3) and the budget constraint (vii), and trivially satisfies the land identity (viii) since $\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i$ holds by construction for any zoning profile. Hence (t^{DE}, I^{DE}) is a feasible point of the planner's problem (30) — one which the planner is free to choose but need not. Since (t^{SP}, I^{SP}) is by definition the maximizer of the same objective $N\Gamma(1 - 1/\epsilon)\Phi^{1/\epsilon}$ over the same feasible set, $W^{SP} \geq W^{DE}$ follows immediately. $\square$

Proposition 2 (The investment margin is welfare-neutral at the margin). To leading order in a neighborhood of the decentralized equilibrium,

$$\Delta W \equiv W^{SP} - W^{DE} \approx \sum_{i \in S} \left. \frac{\partial W}{\partial \bar{H}_i^R} \right|_{DE} (\bar{H}_i^{R, SP} - \bar{H}_i^{R, DE}),$$

i.e. the first-order welfare gap comes entirely from the zoning margin; the investment margin contributes nothing at first order.

Proof. Write $\Delta W \approx \sum_i \left. \frac{\partial W}{\partial \bar{H}_i^R} \right|_{DE} (\bar{H}_i^{R, SP} - \bar{H}_i^{R, DE}) + \sum_{k\ell} \left. \frac{\partial W}{\partial I_{k\ell}} \right|_{DE} (I_{k\ell}^{SP} - I_{k\ell}^{DE})$, the first-order Taylor expansion of W around the DE profile in its two instrument blocks. I^{DE} is

the transportation authority's own optimum given t^{DE} (Definition 1, condition 6), so I^{DE} satisfies the same first-order condition (25) used in §Social Planner's "Why investment does not add a term." By the identical envelope/binding-budget argument — $\partial W/\partial I_{k\ell} = \mu c_{k\ell}$ at I^{DE} , and $\sum_{k\ell} c_{k\ell}(I_{k\ell}^{SP} - I_{k\ell}^{DE}) = K - K = 0$ since both profiles exhaust the same fixed budget K — the second sum is exactly $\mu \sum_{k\ell} c_{k\ell}(I_{k\ell}^{SP} - I_{k\ell}^{DE}) = 0$, leaving only the zoning term. $\square$

Remark. An earlier version of this result (2026-07-21/23) went further, claiming the remaining zoning term itself collapses to a single named "cross-jurisdiction spillover" object, using the municipal first-order condition (16) to zero out the rest. That step does not survive §Social Planner's corrected derivation: (16) is the first-order condition for $W^g = \sum_{i \in S^g} N_i \bar{U}_i$ (municipal, incumbent-style), while $\partial W/\partial \bar{H}_i^R$ here is built from the planner's own $NT(1 - 1/\epsilon)\Phi^{1/\epsilon}$ — a structurally different object (§Social Planner, "Zoning": no channel-(1) analogue, $\bar{U}$ in place of $\bar{U}_i$). Invoking (16) to zero out a quantity it was never the first-order condition for is not valid, so $\partial W/\partial \bar{H}_i^R|_{DE}$ does not vanish and does not reduce to a single clean term. It is, instead, governed by exactly the same $\Lambda_i^\phi, \Lambda_i^q, \Lambda_i^w$ system that closes the planner's own zoning condition (33) (§Social Planner), evaluated at the DE allocation rather than the SP one — deferred to quantification for the same reason $\Lambda_{k\ell}$ and Λ_i are throughout this document. Proposition 1 — the exact, global $W^{SP} \geq W^{DE}$ — does not depend on this and stands on its own; what is not currently available is a closed-form leading-order decomposition of *how* that gap arises beyond "entirely through zoning, not investment" (Proposition 2).

References

- [1] Gabriel M Ahlfeldt et al. "The economics of density: Evidence from the Berlin wall". en. In: *Econometrica: journal of the Econometric Society* 83 (6 2015), pp. 2127–2189.
- [2] Treb Allen and Costas Arkolakis. "The Welfare Effects of Transportation Infrastructure Improvements". en. In: *The Review of Economic Studies* 89 (6 Nov. 7, 2022), pp. 2911–2957.
- [3] Olivia Bordeu. *Commuting Infrastructure in Fragmented Cities*. 2026.
- [4] Pablo D Fajgelbaum and Edouard Schaal. "Optimal transport networks in spatial equilibrium". en. In: *Econometrica: Journal of the Econometric Society* 88 (4 2020), pp. 1411–1452.
- [5] Evan Mast. "Warding off Development: Local Control, Housing Supply, and NIM-BYs". en. In: *The Review of Economics and Statistics* 106 (3 May 14, 2024), pp. 671–680.
- [6] Vincent Rollet and Laura Weiwu. *Measuring Winners and Losers from Increasing Housing Supply*. en. Research rep. 2025.